Al Isra Private University
Faculty of Engineering
Electrical Engineering Department

Student Name:_____
Student Number:_____

Second Semester: 2005/2006
Second Exam

Course Name:	Systems & Signals Processing	Date:	May 7, 2006
Course No:	402-322	ExamTime:	1
Instructor:	Eng. Juma Abu-Ghalune	No. of pages:	1

1. Using Table 4-1 and Table 4-2, find the Fourier Transform of:
   a. $x(t) = 5.exp(-(t+0.1)^2)$.
   b. $y(t) = 4.sin(50\pi t).cos(500\pi t)$.
   c. $z(t) = 4/(1 - j\pi t)^2$.

2. Using Table 4-1 and Table 4-2, find the Fourier inverse transform of:
   a. $X(f) = 4exp(-4f)u(f)$.
   b. $Y(f) = (-j/\pi f)exp(j0.2\pi f)$.

3. The complex exponential Fourier Series over an interval $0 < t < T$ is given by $x(t) = \sum (4/(1+jn))exp(j(10\pi nt - n\pi/9))$ for all n. Find:
   a. The fundamental frequency.
   b. The average value.
   c. The amplitude and the phase of the third harmonic.
   d. The average power contained in the first three harmonics.

4. Consider the signal $x(t) = cos(100\pi t)sin(200\pi t) + cos^2(150\pi t)$. Find:
   a. The fundamental frequency.
   b. The coefficients a_n, b_n, and X_n for the signal.
   c. The average power of $x(t)$.

5. Using the integral form, find the Fourier transform of the signal $x(t) = 4.exp(-2.t.sgn(t))$.

Scanned by CamScanner

Q1 a. $x(t) = 5e^{-(t+0.1)^2} = 5e^{-\pi\left(\frac{t}{\sqrt{\pi}}+0.1\right)^2}$

$X(f) = \frac{5}{(\sqrt{\pi})^{-1}} e^{j2\pi(0.1)f} e^{-\pi\left(\frac{\pi}{\sqrt{\pi}}f\right)^2}$

$= 5\sqrt{\pi}\, e^{j0.2\pi f} e^{-\pi^2 f^2}$

b. $y(t) = 4\sin(50\pi t)\cos(500\pi t) - 2\sin(550\pi t) + 2\sin(450\pi t)$

$Y(f) = j[\delta(f+275) - \delta(f-275)] - j[\delta(f+225) - \delta(f-225)]$

c. $z(t) = \frac{4}{(1-j\pi t)^2} = \frac{4\times 4}{(2+j2\pi(-t))^2}$

$Z(f) = 16 f e^{-2f} u(f)$

Q2 a. $X(f) = 4e^{-4f}u(f) =$

$x(t) = \frac{4}{4 - j2\pi t}$

b. $Y(f) = \left(\frac{-j}{\pi f}\right) e^{j0.2\pi f} = \left(\frac{-j}{\pi f}\right) e^{j2\pi(0.1)f}$

$y(t) = \text{sgn}(t+0.1)$

Q3 $x(t) = \sum \frac{4}{1+jn} e^{-jn\pi/9} e^{j2\pi(5)nt}$

$X_n = \frac{4}{1+jn} e^{-jn\pi/9}$

a. $f_0 = 5$

b. $X_0 = 4$ the average value

Scanned by CamScanner

Continue

Q.3 c. $X_3 = \frac{4}{1+j3} e^{-j\frac{\pi}{3}} = \frac{4}{\sqrt{10}} e^{-j\left(\frac{\pi}{3} + \tan^{-1}(3)\right)}$

d.

$|X_1| = \frac{4}{\sqrt{2}}$, $|X_2| = \frac{4}{\sqrt{5}}$, $|X_3| = \frac{4}{\sqrt{10}}$.

$P_{avg} = |X_0|^2 + 2\left[|X_1|^2 + |X_2|^2 + |X_3|^2\right]$.

$= 16 + 2\left[\frac{16}{2} + \frac{16}{5} + \frac{16}{10}\right] = 16 + 32\left[5 + 2 + 1\right]\frac{1}{10}$

$= 16 + 32 \times \frac{8}{10}$

Q.4 $x(t) = \cos(100\pi t)\sin(200\pi t) + \cos^2(150\pi t)$

$= \frac{1}{2} + \frac{1}{2}\sin(100\pi t) + \frac{1}{2}\cos(300\pi t) + \frac{1}{2}\sin(300\pi t)$

a. $f = 50$ Hz

b. $a_0 = \frac{1}{2}$, $b_1 = \frac{1}{2}$, $a_3 = \frac{1}{2}$, $b_3 = \frac{1}{2}$

$X_0 = \frac{1}{2}$, $X_1 = -j\frac{1}{4}$, $X_3 = \frac{1}{4} - j\frac{1}{4}$.

c. $P_{avg} = |X_0|^2 + 2\left[|X|^2 + |X_3|^2\right] = \frac{1}{4} + 2\left[\frac{1}{16} + \frac{1}{16} + \frac{1}{16}\right]$

$= \frac{10}{16} = 1.75$ W

Q.5 $x(t) = 4e^{-2t\,\mathrm{sgn}(t)} = \begin{cases} 4e^{2t} & t < 0 \\ 4e^{-2t} & t > 0 \end{cases}$

$X(f) = \int_{-\infty}^{0} 4e^{(2-j2\pi f)t}\,dt + \int_{0}^{\infty} 4e^{-(2+j2\pi f)t}\,dt$

$= \frac{+4}{(2-j2\pi f)} + \frac{4}{(2+j2\pi f)} = 4 \cdot \frac{4}{4 + (2\pi f)^2}$

Scanned by CamScanner